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[4657]-575

S.E. (Computer Engg.) (First Semester) EXAMINATION, 2014

MICROPROCESSOR ARCHITECTURE

(2012 PATTERN)

Time : Two Hours

Maximum Marks : 50

N.B. :— (i) Attempt Q. No. 1 or Q. No. 2, Q. No. 3 or Q. No. 4,
Q. No. 5 or Q. No. 6, Q. No. 7 or Q. No. 8.

(ii) Neat diagrams must be drawn wherever necessary.

(iii) Figures to the right indicate full marks.

(iv) Assume suitable data, if necessary.

1. (a) Explain the following pins of 8086 : [6]

(i) $\overline{\text{BHE}}$

(ii) ALE

(iii) $\text{MN}/\overline{\text{MX}}$

(b) Explain flag register of 80386. [4]

(c) What is the difference between GDT and LDT ? [3]

P.T.O.

Or

2. (a) What are the functions of the following pins : [6]
- (i) NMI
 - (ii) ERROR#
 - (iii) $\overline{\text{BE}}_0 - \overline{\text{BE}}_3$.
- (b) Explain memory organization of 8086. [4]
- (c) How is logical address converted to physical address in 8086 ? [3]
3. (a) What are characteristics of address pipelining in 80386 ? [4]
- (b) Explain timing diagram for pipelined write cycle. [5]
- (c) Explain macro in assembly language programming with syntax and example. [3]

Or

4. (a) Explain the following instructions of 80386 : [5]
- (i) SLDT
 - (ii) LEA
 - (iii) OUT
 - (iv) CMP
 - (v) XCHG.

- (b) Explain timing diagram for read cycle in non-pipelined mode. [4]
- (c) Enlist and explain any *three* addressing modes of 80386. [3]
5. (a) What are the *three* common configurations that support multiprocessing ? Explain. [6]
- (b) What is single instruction multiple data model for parallel processing ? [4]
- (c) Define CMP. [3]

Or

6. (a) What are the advantages of multicore designing ? [3]
- (b) What are the advantages of cache memory ? [4]
- (c) What are the different architectures of multicore ? Explain. [6]
7. (a) Explain hyperthreading with advantages and disadvantages. [5]
- (b) Give the features of SSE. [4]
- (c) Enlist data types of 64-bit architecture. [3]

Or

8. (a) Explain virtualization technology. [5]
- (b) What are the operating modes of Intel 64-bit architecture ? [4]
- (c) What are the differences between IA-32 bit Basic Execution Environment and IA-64 bit Basic Execution Environment ? [3]